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INDR 252 CLASS EXERCISE, April 01, 2024

A cigarette manufacturer claims that the average nicotine content μ of brand B cigarettes is at most 1.5 mg. Rejecting the manufacturer's claim without strong, contradictory evidence would be unwise. Suppose the nicotine content is known to be normal with $\sigma=0.2$. Assume $n=10$.

1. Formulate the hypothesis testing.
2. Find the rejection region for $\alpha = 0.05$, i.e., an appropriate bound for $\bar{X}$. Do you reject H_0 if $\bar{x} = 1.596$? Why or why not? $z_{0.05}=1.645$
3. If you cannot reject H_0 , what is the probability of a type-II error if $\mu = 1.6$, i.e., $\beta(1.6)$?

$$1. H_0: \mu = \mu_0 = 1.5 \quad (\mu \leq \mu_0)$$

$$H_a: \mu > \mu_0 = 1.5$$

$$3. \beta(1.6) =$$

$P(\text{cannot reject } H_0, H_0 \text{ is true})$

$$2. \alpha = 0.05 = P(\text{reject } H_0, H_0 \text{ is true})$$

$$= P(\bar{X} > h, \mu = 1.5) = P\left(\frac{\bar{X} - 1.5}{0.2/\sqrt{10}} > \frac{h - 1.5}{0.2/\sqrt{10}}\right)$$

$$= P(\bar{X} < h, \mu = 1.6)$$

$$= P(Z > z_\alpha) \rightarrow \frac{h - 1.5}{0.2/\sqrt{10}} = z_{0.05} = 1.645$$

$$= P\left(Z < \frac{h - 1.6}{0.2/\sqrt{10}}\right)$$

z	0.00	0.01	0.02	0.03	0.04	0.05
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678

$$= P\left(Z < \frac{1.604 - 1.6}{0.2/\sqrt{10}}\right)$$

$$= P(Z < 0.064)$$

$$\approx 0.52$$

$\rightarrow h = 1.604 \nless \bar{x} = 1.596 \rightarrow \text{cannot reject } H_0 \text{ at } \alpha = 5\%$